

System Architecture Specification Document

AI-Powered Automatic Block Planning System (Smart India Hackathon PS 26027)

1. Architectural Overview & Vision

The Indian Railways Automatic Block Planning System replaces decentralized, manual corridor block requests with an AI-driven, multi-departmental constraint optimization engine. It integrates live maintenance defects from TMS (Track), SMMS (Signals), and TDMS (Traction) with passenger train timetables and Control Office freight forecasts.

INDIAN RAILWAYS AUTOMATIC BLOCK PLANNING SYSTEM ARCHITECTURE



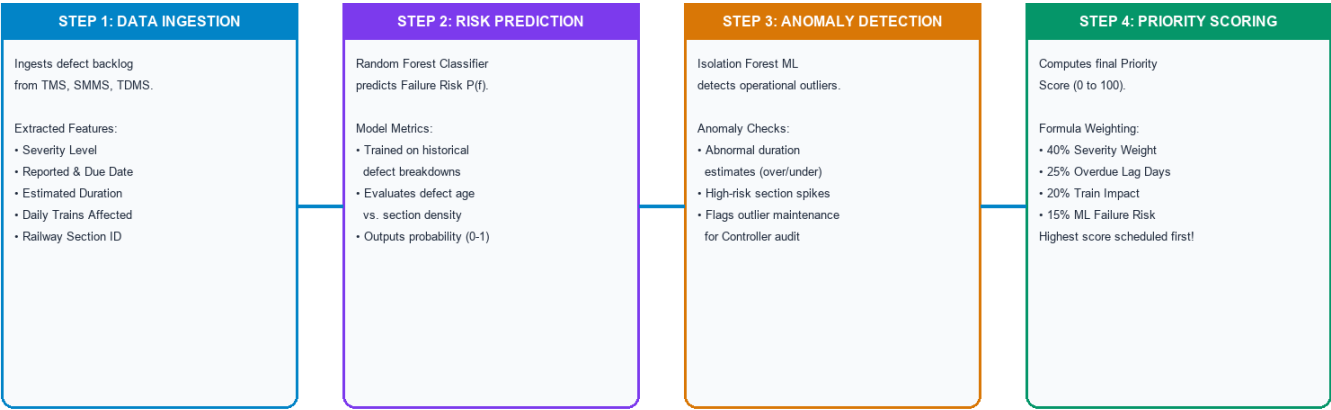
Key Architectural Layers:

- Data Layer: Unified SQLite schema ingesting 6,300+ defects, 2,400 corridor slots, 2,000 timetables, and 2,000 freight forecasts.
- AI/ML Layer: Random Forest failure risk prediction + Isolation Forest anomaly detection + mathematical priority scoring.
- Optimization & Agent Layer: Google OR-Tools CP-SAT constraint solver + Multi-Department Shadow Block Clustering + 13 Governance Agents.
- Presentation Layer: Streamlit dual-panel workspace with Plotly zoomable interactive charts, automated PDF reporting, and Groq-powered voice AI assistant.

2. AI/ML Prioritization & Mathematical Scoring

Maintenance tasks are ranked dynamically using a combined mathematical and Machine Learning scoring model. Defects are evaluated based on structural severity, overdue lag days past deadline, daily train impact, and ML-predicted failure probability.

AI/ML DEFECT PRIORITIZATION & RISK SCORING PIPELINE



Prioritization Formula:

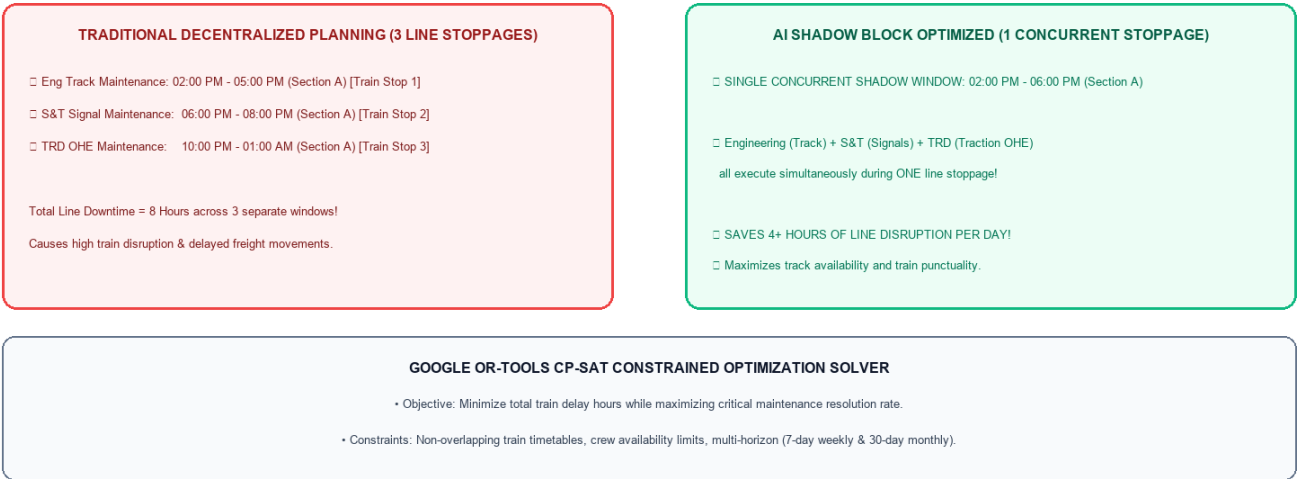
Priority Score = (Severity_Weight * 0.40) + (Overdue_Lag_Days * 0.25) + (Trains_Affected * 0.20) + (ML_Failure_Risk * 0.15)

Where Severity Weights are: Critical = 100, High = 75, Medium = 50, Low = 25. Overdue lag represents actual days delayed past deadline.

3. Google OR-Tools Constraint Optimization & Shadow Blocks

The optimization core uses Google OR-Tools CP-SAT to solve multi-horizon corridor block schedules (7-day Weekly tactical and 30-day Monthly strategic plans). A key innovation is Shadow Block Clustering, which groups Engineering, S&T, and TRD tasks into single concurrent maintenance windows.

MULTI-DEPARTMENT SHADOW BLOCK CLUSTERING OPTIMIZATION



4. Multi-Agent Architecture & Governance Tiers

The system deploys 13 specialized autonomous agents that coordinate tasks, monitor safety SLAs, and execute auto-approvals:

1. Department Agent: Filters and manages department-specific defects.
2. Traffic Agent: Ensures train timetable non-overlap and passenger advisories.
3. Coordinator Agent: Orchestrates cross-department shadow block clustering.
4. Replanning Agent: Manages real-time disruptions using 3 Risk Tiers:
 - Tier 1 (Routine): Auto-approves minor gap-filling.
 - Tier 2 (Medium Risk): Auto-approves and logs controller alert.
 - Tier 3 (Critical Risk): Escalates to manual Controller override.
5. Feedback Loop Agent: Records execution minutes and computes Early/On-Time/Late performance variance.
6. Compliance Agent: Enforces max block duration limits and safety regulations.

5. AI Assistant & Multilingual Voice Architecture

The application integrates a dedicated, persistent AI Assistant powered by Groq API (Qwen 27B model):

- Speech Recognition: Web Speech API transcribes English, Hindi, and Telugu in real-time.
- Direct SQL NLP Querying: Understands natural time queries like '2pm to 3pm what scheduled' and defect lookups ('TMS-00010').
- Automated Explainer: Explains the 5-step report generation process on demand.
- Unified UI: Single rounded prompt container combining voice mic and text send button.

6. Security & Auditability

- Role-Based Access Control (RBAC): Engineering, S&T, TRD, Admin.
- Password Security: BCrypt password hashing.
- Audit Logging: All controller overrides and agent decisions recorded in audit_log table with timestamps.